

Comparisons

Session 8

ISS Data Visualization with R
Faculty of International Social Sciences, Gakushuin University
Spring 2026

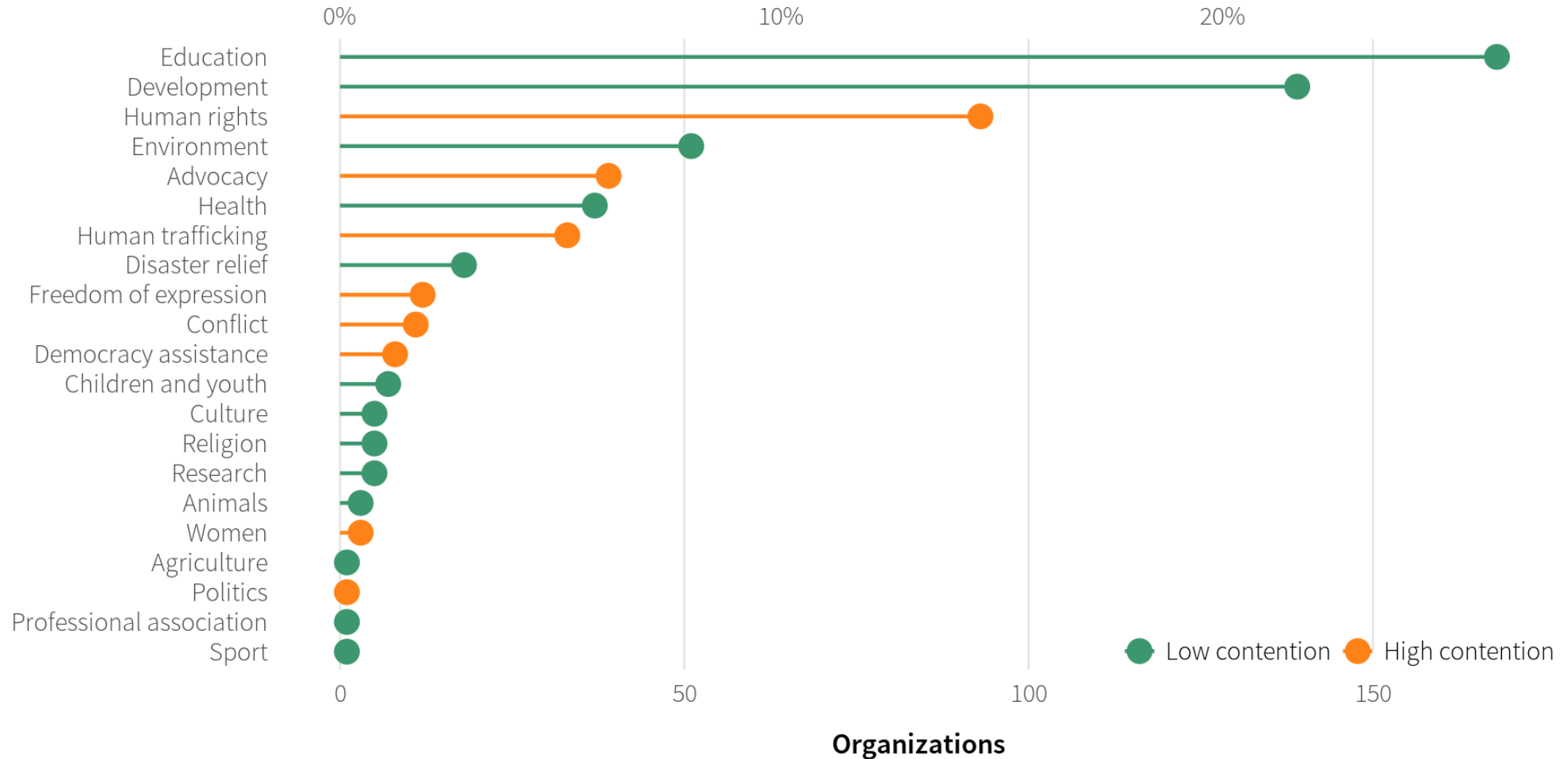
Plan for today

Visualizing comparisons

Reproducible examples

Visualizing comparisons

Lollipops and bars





Small multiples

How Trump compares with past presidents

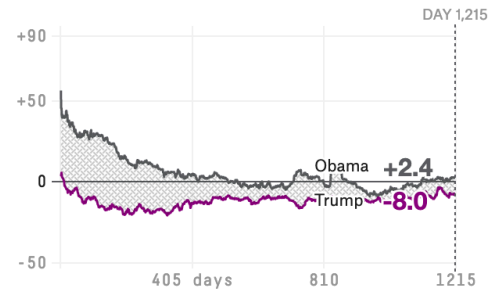
○ Approval rating ○ Disapproval rating ● Net approval

1,215 days

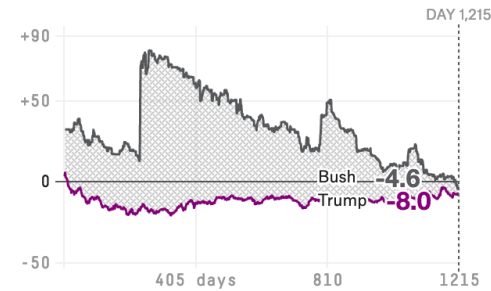
4 years

8 years

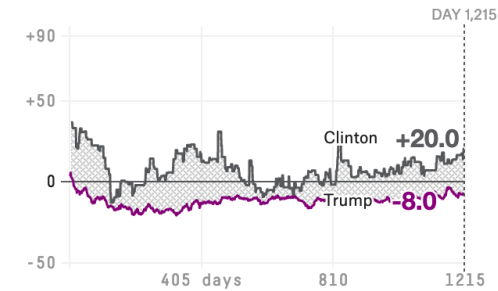
Barack Obama 2009-17



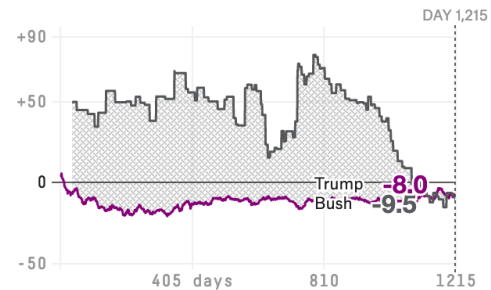
George W. Bush 2001-09



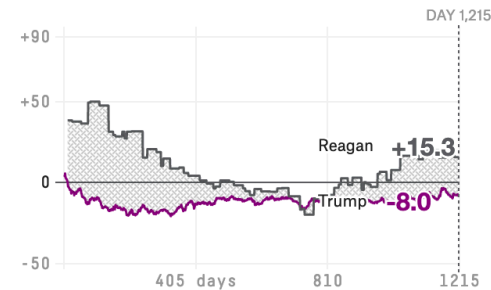
Bill Clinton 1993-2001



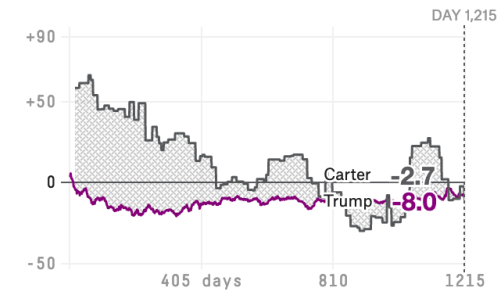
George H.W. Bush 1989-93



Ronald Reagan 1981-89

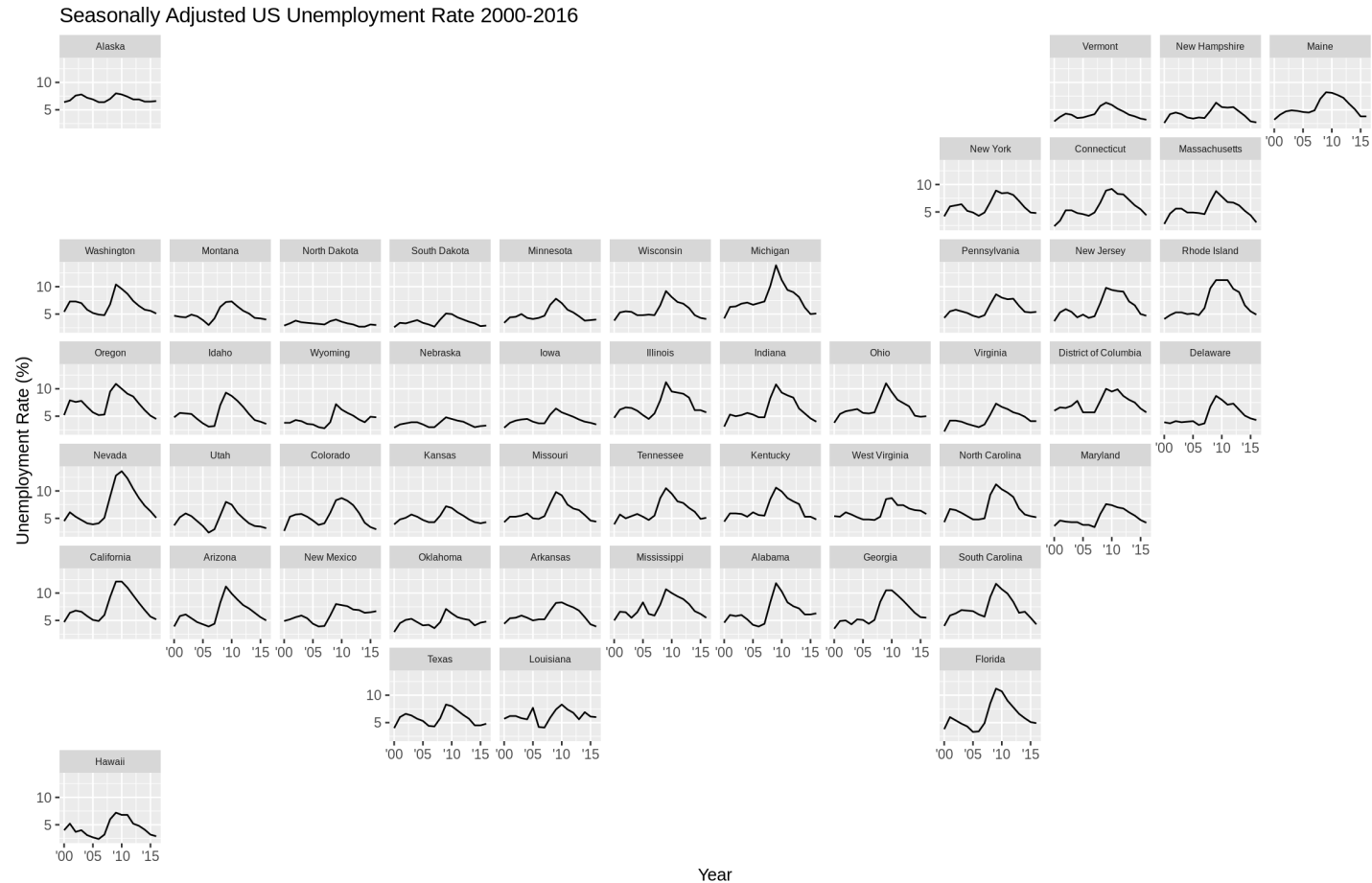


Jimmy Carter 1977-81



FiveThirtyEight, Trump approval ratings

Small multiples with larger shapes



Data Source: bls.gov

`facet_geo()` in the **geofacet** package

Sparklines

Mauricio Pochettino has lead Spurs on their best run **8TH**  **2ND** in 24 years of the Premier League

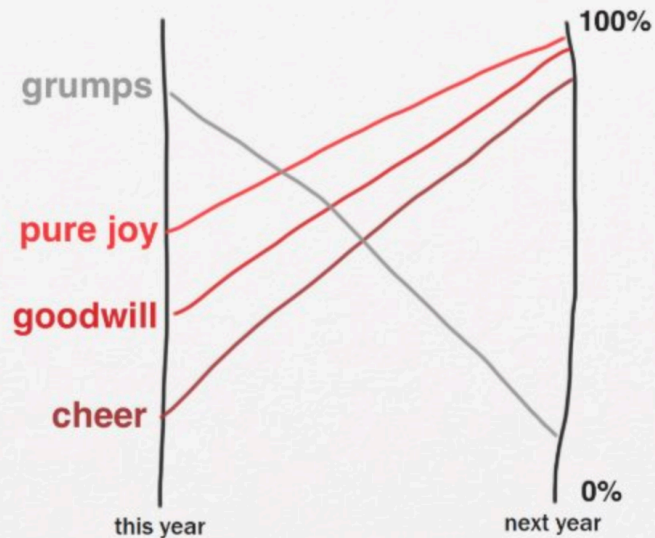
Alibaba stock is at 5 yr high **93.89**  **152.11** as of July 2017

The FTSE100 Brexit bounce **5562**  **7501** continues one year on from the vote last summer



Slopegraphs

**May your New Year see
a significant increase in
all the good stuff.**



Slopegraphs

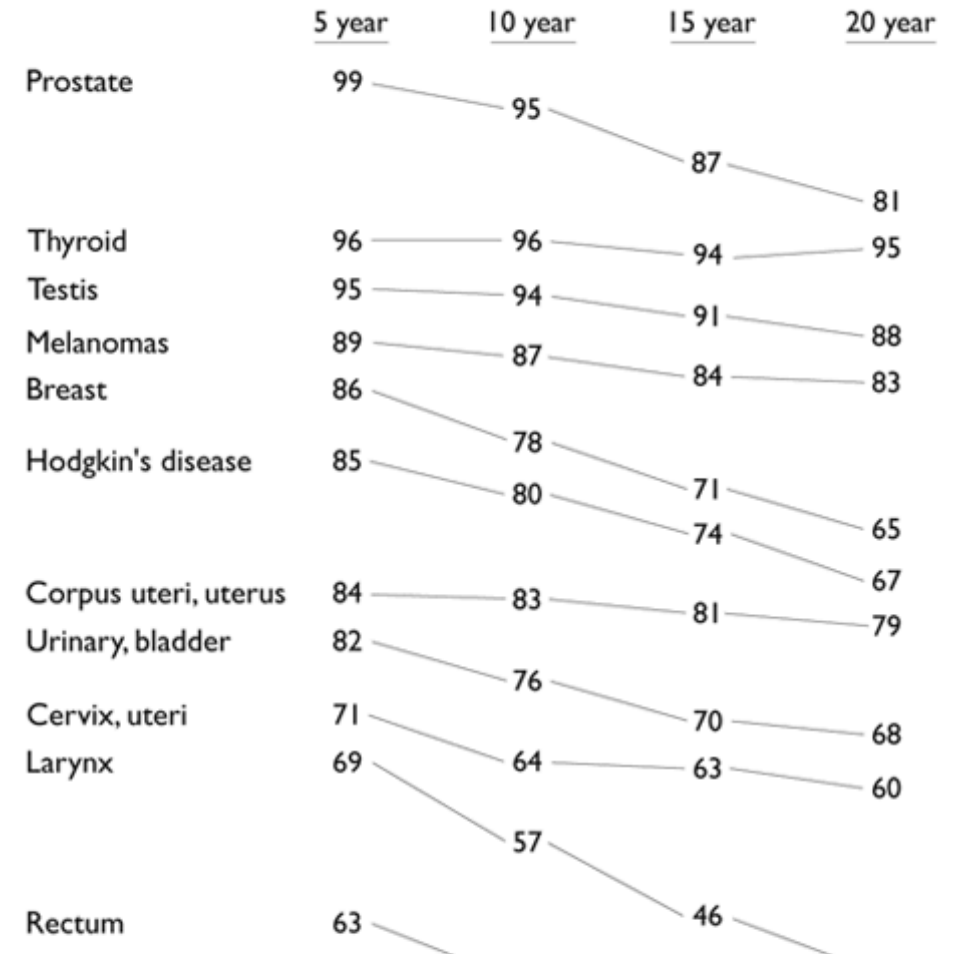


Slopegraphs

Estimates of relative survival rates, by cancer site

	% survival rates and their standard errors							
	5 year		10 year		15 year		20 year	
Prostate	98.8	0.4	95.2	0.9	87.1	1.7	81.1	3.0
Thyroid	96.0	0.8	95.8	1.2	94.0	1.6	95.4	2.1
Testis	94.7	1.1	94.0	1.3	91.1	1.8	88.2	2.3
Melanomas	89.0	0.8	86.7	1.1	83.5	1.5	82.8	1.9
Breast	86.4	0.4	78.3	0.6	71.3	0.7	65.0	1.0
Hodgkin's disease	85.1	1.7	79.8	2.0	73.8	2.4	67.1	2.8
Corpus uteri, uterus	84.3	1.0	83.2	1.3	80.8	1.7	79.2	2.0
Urinary, bladder	82.1	1.0	76.2	1.4	70.3	1.9	67.9	2.4
Cervix, uteri	70.5	1.6	64.1	1.8	62.8	2.1	60.0	2.4
Larynx	68.8	2.1	56.7	2.5	45.8	2.8	37.8	3.1
Rectum	62.6	1.2	55.2	1.4	51.8	1.8	49.2	2.3
Kidney, renal pelvis	61.8	1.3	54.4	1.6	49.8	2.0	47.3	2.6
Colon	61.7	0.8	55.4	1.0	53.9	1.2	52.3	1.6
Non-Hodgkin's	57.8	1.0	46.3	1.2	38.3	1.4	34.3	1.7
Oral cavity, pharynx	56.7	1.3	44.2	1.4	37.5	1.6	33.0	1.8
Ovary	55.0	1.3	49.3	1.6	49.9	1.9	49.6	2.4
Leukemia	42.5	1.2	32.4	1.3	29.7	1.5	26.2	1.7
Brain, nervous system	32.0	1.4	29.2	1.5	27.6	1.6	26.1	1.9

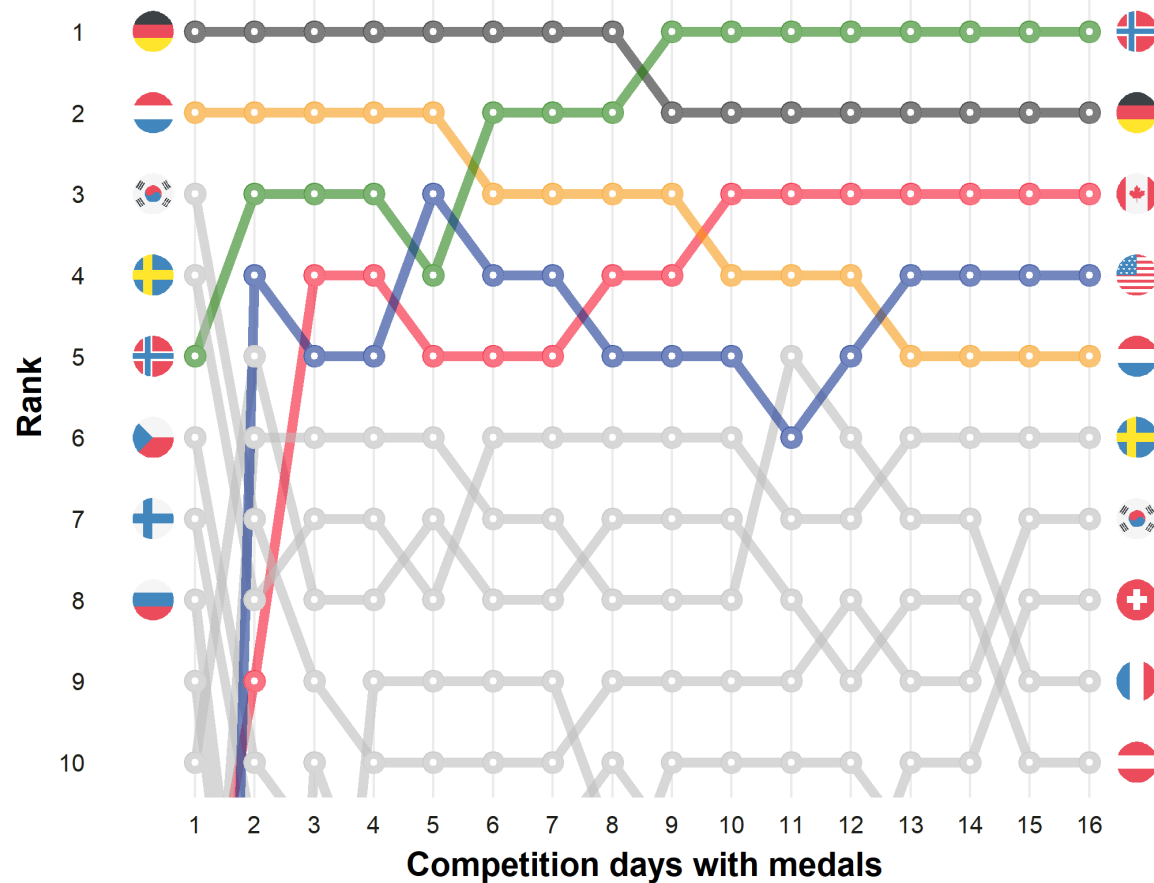
Estimates of % survival rates



Bump charts

PyeongChang 2018 Olympic Winter Games

Countries ranked by overall medals after each competition day



Reproducible examples

This is 100% normal!



Brandon Rohrer

@_brohrer_



I was just asked whether I ever have to look things up when I code. I want to go on record saying that, aside from canned white board coding examples, I can't write two lines of code without referring to Google or stack overflow. I would be lost without them.

3:57 PM · Oct 8, 2018 · [Twitter for iPhone](#)

298 Retweets **1.5K** Likes



Broken cake



Help! My cake broke!

vs.

**Help! I followed these
6 steps and my cake broke!**

**Same principle
applies to code**

Reprexes

Reproducible examples

**Something anyone can run on their computer
to reproduce the problem you're facing**

Debugging and replexes

Simplify your code down to something very basic

Add additional things until stuff breaks

Use a subset of your data or invent fake data

Restart your session and see if it runs in a new session

Ask the internet for help using your toy example

75% of the time you'll find what's wrong as you make the replex!

Making datasets with `tribble()`

`.pull-right[`

```
my_data <- tribble(  
  ~animal, ~number,  
  "cat", 5,  
  "dog", 4,  
  "bear", 7,  
  "bison", 1  
)
```

`my_data`

```
## [38;5;250m3 [39m bear      7  
## [38;5;250m4 [39m bison     1
```

```
## [38;5;246m# A tibble: 4 × 2 [39m  
##   animal number  
##   [3m [38;5;246m<chr> [39m [23m  
## [38;5;250m1 [39m cat           5  
## [38;5;250m2 [39m dog           4
```

Example reprex

```
my_data <- tribble(  
  ~animal, ~number,  
  "cat", 5,  
  "dog", 4,  
  "bear", 7,  
  "bison", 1  
)
```

```
# This plot has a fill legend, but I want to remove it because it's redundant  
# What's the best way to get rid of the fill?
```

```
ggplot(fake_data, aes(x = animal, y = number, fill = animal)) +  
  geom_col()  
# I add something here, but what?
```